

Oikos

System design summary: home health intake agent

Event	Healthcare AI Hackathon (AWS, San Francisco), 26 September 2026
Team	Sankalpa Venkatraghavan, Terry Lin, Himani Shah
Repository	github.com/sankalpav/Oikos
Stack	Python, Streamlit, GPT-5.6 Terra on Amazon Bedrock (OpenAI-compatible Responses API), pandas
Data	100% synthetic. No real patient information was used at any point.
Status	Working demo; live model verified end to end; cached fallback for offline use
Version	v2: reflects the team's latest code (discharge-planner note, PDGM estimate, multi-state programs) and corrected statistics

One line: every home health referral gets a next step. Accepted patients are billed fully; declined referrals go back to the hospital with a note of next steps.

Why the name: *oikos* is Greek for household or home, the place where both home health and home care happen.

1. Executive summary

Hospitals and skilled nursing facilities (SNFs) send discharge referrals to home health agencies, mostly through CarePort, fax and phone. Agencies screen these by hand, respond slowly, and now accept only about a quarter of home health referrals, driven by staffing challenges and rising patient complexity. Of the patients they do accept, the coder sees only a short intake-nurse summary, so documented comorbidities never reach the claim. Patients they decline receive no next step, even when they qualify for non-medical home care programs.

Oikos is an intake agent that an agency subscribes to. For each referral it (1) extracts clinical facts from the discharge packet with an LLM, (2) applies deterministic rules for insurance, eligibility, paperwork and nurse capacity to decide accept / fixable / decline, then (3a) for accepted patients surfaces missed billable diagnoses with chart evidence, or (3b) for declined referrals checks home care programs and adds a note to the referral file for the hospital's discharge planner. Oikos never contacts patients or families; only the hospital does.

The key strategic decision was that **nurses, not patients, are the bottleneck**. Oikos is therefore positioned as maximizing revenue per nurse and giving every referral an outcome, rather than simply generating more referrals.

2. Problem context

2.1 How a referral flows today

- A hospital or SNF case manager creates a discharge referral and broadcasts it to many agencies via CarePort, fax or phone.
- An agency intake coordinator reads the packet by hand and checks insurance, homebound status, skilled need and the face-to-face encounter documentation.
- The coordinator checks whether a qualified clinician can reach the patient's zip code within the 48-hour start-of-care window required by Medicare.

- The first agency to accept usually wins the patient (WellSky reports first responders are ~22% more likely to secure the placement; company-reported).
- An RN performs the start-of-care visit and OASIS assessment, writes a short summary, and sends it to a coder, who bills a 30-day PDGM payment period using only that summary.
- Declined referrals are dropped; the patient goes elsewhere or receives nothing.

2.2 Market and bottleneck statistics used

Statistic	Value	Source	Confidence
Medicare-certified home health agencies	12,000+	MedPAC, Dec 2025 (2024 data)	High
Traditional Medicare home health users / yr	2.7M	MedPAC, Dec 2025	High
30-day payment periods / yr	8.3M	MedPAC, Dec 2025	High
Traditional Medicare home health spend / yr	\$16.0B	MedPAC, Dec 2025	High
Medicare margin vs all-payer margin	21.2% vs 5.0%	MedPAC, Dec 2025	High
Home health referral acceptance	~24% (Dec 2022), from ~46% (2019)	WellSky 2023 Evolution of Care Report (CarePort network data), via Home Health Care News	High
Home health referrals, year over year	+4.6%	WellSky network data via Home Health Care News, Dec 2025	High
Comorbidity burden (Van Walraven) since 2019	+34%	WellSky via Home Health Care News, Dec 2025	High
First responders more likely to secure placement	+22%	WellSky, Mar 2026 (company-reported, no published methodology)	Low - verify in pilot
Initial assessment deadline	48 hours	42 CFR 484.55(a)(1): within 48 hours of referral or return home	High

Removed in v2: a '77% of providers cite staffing' figure (traced to a 2023 direct-support/IDD workforce survey, not home health) and a '~30% from ~80%' acceptance figure from a vendor blog. Both were replaced with first-party or regulatory sources.

2.3 Home health vs home care

	Home health (skilled)	Home care (non-medical)
Services	Nursing, PT/OT/SLP, wound care, IV, medication management	Bathing, dressing, meals, housekeeping, companionship
Order required	Yes: physician certifies, plan of care	No
Eligibility	Homebound + intermittent skilled need	Anyone who needs help
Main payers	Medicare (FFS and Advantage), Medicaid, commercial	Private pay, LTC insurance, Medicaid waivers / IHSS, VA
Role in Oikos	The paying customer's core business	Fallback route for declined patients

3. Key decisions and rationale

Decision	Options considered	Choice	Rationale
Bottleneck to solve	More patients vs more nurse capacity	Nurse capacity; maximize revenue per nurse	Agencies already decline about three in four referrals, driven by staffing and patient complexity. More referrals alone does not add revenue.
Customer	Home health agencies (B2B), families (D2C), hospitals	Sell to agencies (B2B)	Agencies have budget and a clear ROI. D2C has high acquisition cost, and agencies may not contact declined patients directly.
Patient contact	Text families a care plan vs note on the referral file	Note to the hospital discharge planner on the referral file only	Agencies may not contact declined patients; only the referring hospital communicates with the patient.
Integration path	Integrate with CarePort directly, or act on the agency's behalf	Agent runs on each agency's own portal access	CarePort requires an NPI and has no preference logic (first come, first served); an agency-side agent avoids needing a partnership.
Revenue directions	1) reroute rejected patients, 2) source from SNFs/LTC, 3) billing and coding	Demo builds 1 and 3 on top of triage; 2 on the roadmap	SNF sourcing relies on phone/fax relationships and is not demoable in an hour; the other two are self-contained.
Hackathon scope	Triage only, triage + one layer, all three	All three layers (team choice)	Shows the full story: every referral gets an outcome.
LLM vs rules split	End-to-end LLM decisioning vs LLM extraction + deterministic rules	LLM for reading and writing; rules for every decision	Decisions must be auditable and repeatable; eligibility and staffing are rule-shaped; LLMs are strong at unstructured text.
Model	Claude via Bedrock, GPT-5.6 Terra via Bedrock	GPT-5.6 Terra (team choice)	Matches the AWS workshop setup already configured; OpenAI-compatible Responses API.
Resilience	Live only vs live with fallback	Live with per-step cached fallback	Workshop credentials expire in ~24 hours and demos cannot depend on Wi-Fi.
UI	Streamlit, Next.js, CLI	Streamlit	Fastest path to a demoable multi-view app in Python; free public hosting.
Billing safety	Auto-code vs surface evidence	Surface evidence only; clinician confirms, coder codes	Avoids upcoding risk; keeps clinical judgment with the nurse.
First market	National vs one state	California first; programs table covers 6 states	IHSS pays family members as caregivers; team is local. State Medicaid programs live in a data table so new states need no code changes.
Billing dollar estimate	LLM-estimated dollars vs rule-based PDGM math	Rule-based estimate, Original Medicare only	Dollar figures must be deterministic and auditable; PDGM only prices Medicare fee-for-service periods.
Statistics	Vendor blog figures vs first-party and regulatory sources	MedPAC, WellSky network data, CFR	Two vendor-blog figures turned out to be mis-sourced; every number now traces to a primary source with its confidence stated.

Decision	Options considered	Choice	Rationale
Pricing	Per-referral or percentage-of-revenue vs flat fee	Flat per-branch software fee (+ private-pay home care referral fees)	Referral-linked payments raise federal anti-kickback concerns, especially for Medicaid/IHSS-funded patients.
Hosting	GitHub Pages, Streamlit Community Cloud, laptop tunnel	Streamlit Community Cloud from the GitHub repo	GitHub Pages is static only; Community Cloud is free and runs Python. Public link runs cached mode so no one can spend AWS credits.
Name	Constant Care (placeholder), Baton, Census, Homebound, others	Oikos	Team choice; means household or home.

4. System architecture

4.1 Pipeline

```

Referral (structured header + discharge packet text)
|
v
[1] LLM extraction -----> diagnoses + ICD-10 + exact quotes, homebound, skilled need,
|                           services (discipline + skills), face-to-face, ADL/IADL needs
|
v
[2] Rules: triage -----> insurance | skilled need | homebound | face-to-face | staffing
|                               => ACCEPT / FIXABLE / DECLINE (+ reason)
+-- ACCEPT / FIXABLE --> [3a] LLM billing capture (missed dx vs nurse summary)
|                       + rules: PDGM comorbidity dollar estimate (Original Medicare)
|                       + templated fix actions (face-to-face request, prior auth)
+-- DECLINE -----> [3b] Rules: home care qualification + agency match
|                       + LLM note on the referral file (for the hospital discharge planner)
v
Quote verification: every evidence quote is checked verbatim against the source text

```

4.2 Components

File	Responsibility
app.py	Streamlit UI: Overview (problem statistics), Referral queue, Referral detail, Impact (metrics and ROI sliders). Sidebar toggles live model, tests the Bedrock connection, and runs all referrals in parallel (5 threads).
pipeline.py	Prompts and schemas, triage rules, nurse matching, fix actions, home care rules, orchestration, quote verification, and mock fallback per step.
llm.py	Bedrock client using the OpenAI SDK against the Bedrock OpenAI-compatible endpoint. Uses OPENAI_API_KEY if set, otherwise generates a short-lived Bedrock token from standard AWS credentials. Robust JSON parsing.
data/	Synthetic referrals, nurse roster, home care agencies, payer table, zip centroids (with state), PACE areas, state Medicaid home care programs, PDGM comorbidity subgroups, cached model outputs.

4.3 LLM steps

Step	Input	Output (JSON)	Guardrails in prompt
Extraction	Discharge packet; allowed skill vocabulary; allowed ADL/IADL list	summary, primary_diagnosis, diagnoses[], homebound, skilled_need, services[], face_to_face_documented, adl_needs[], cognitive_impairment	Use only stated facts; every evidence field is an exact quote; home health aftercare/sequela ICD-10 conventions; at most 2 essential skills per service
Billing capture	Discharge summary, nurse summary, extracted diagnoses	missed[] (name, icd10, evidence, why_it_matters), coding_notes[], nurse_query	Exclude conditions already mentioned in the nurse summary, even by abbreviation; apply ICD-10 'with' conventions; never invent; clinician confirms
Discharge-planner note	Home care options, matched agencies, skilled follow-up flag, family's preferred language and contact on file	note (markdown, numbered next steps)	Addressed to the discharge planner and never sent to the patient; leads with home care options, then skilled follow-up; flags preferred language for interpreters; 'likely' / 'may qualify' only

4.4 Triage rules

Check	Pass	Fixable	Fail (decline)
Insurance	Contracted payer, no prior auth	Contracted payer requiring prior auth (drafts request)	Payer not contracted / out of network
Skilled need	Documented skilled nursing or therapy need	-	No skilled need
Homebound	Documented taxing effort / assistance to leave home	-	Not homebound
Face-to-face	Encounter documented	Missing (drafts request to discharge planner, holds slot 4 hours)	-
Staffing	Every required discipline covered in time	-	Any discipline uncovered

Decision logic: any **fail** gives DECLINE (reason = first failing check); otherwise any **fixable** gives FIXABLE; otherwise ACCEPT.

4.5 Nurse matching algorithm

- For each required service, candidates are clinicians whose license equals the discipline (RN, PT, OT, SLP). LVNs cannot open a case.
- The start-of-care discipline is RN if ordered, otherwise the first therapy; it must be seen within 48 hours of discharge. Other disciplines must be seen within 5 days.
- A clinician qualifies if they have all required skills and the patient is within their travel radius (haversine distance between zip centroids).
- Among qualified clinicians, the earliest open slot before the deadline is chosen. If none meets the deadline, the nearest qualified clinician and their first slot are shown as the reason for declining.

4.6 Home care qualification rules (general program rules)

State Medicaid in-home programs come from data/state_programs.json, keyed by the patient's state. Each entry lists eligibility flags (Medicaid required, minimum age, any need or high need) that a generic rule checks.

State	Medicaid in-home programs in the table
CA	IHSS (family member can be the paid caregiver); Medi-Cal MSSP care management
TX	STAR+PLUS HCBS (Personal Assistance Services)
FL	Statewide Medicaid Managed Care, Long-Term Care (SMMC-LTC)
NY	CDPAP (Consumer Directed Personal Assistance Program)
IL	Community Care Program (CCP)
PA	Community HealthChoices (CHC)
Other states	Generic state Medicaid Home and Community-Based Services (HCBS) entry

Program	Rule used in demo	Status shown
VA Aid & Attendance	Wartime veteran or surviving spouse, and at least one ADL need	Possible (income and asset limits apply)
PACE	Age 55+, zip in a PACE service area, and 2+ ADLs or cognitive impairment	Possible
Long-term care insurance	Has a policy, and 2+ ADLs or cognitive impairment (common benefit trigger)	Likely
Private pay	Agencies serving the zip; hours/week = 4 per ADL + 2 per IADL (cap 40); cost = rate x hours x 4.33	Always listed

If the patient still needs skilled care and was declined for staffing or insurance, Oikos adds a warning to notify the discharge planner immediately so another agency can accept the skilled referral. Home care never replaces skilled care.

4.7 PDGM comorbidity dollar estimate

- Runs only for Original Medicare (fee-for-service) patients; PDGM does not price Medicare Advantage or commercial claims.
- Missed ICD-10 codes are matched by prefix to data/comorbidity_subgroups.json (12 curated subgroups, 1 high-tier interaction pair).
- Any matched subgroup gives the low comorbidity tier (+6.01% case-mix); a matched interaction pair gives the high tier (+18.96% total).
- Dollar impact per 30-day period = CY2026 national base rate (\$2,038.22, CMS-1828-F) x tier percentage; per admission = x 1.5 periods.
- Illustrative only: CMS's real list has about 20 low subgroups and about 98 interaction pairs, recalibrated every year. Z-codes and stage 2 pressure injuries are deliberately not modeled.

4.8 Impact and ROI model

- Average payment per 30-day period = \$16.0B / 8.3M periods = about **\$1,928** (MedPAC 2024 data).
- Periods per admission = **1.5** (assumption).
- Recovered admissions / month = referrals x (1 - acceptance rate) x share of declines that are insurance-eligible x recovery rate.
- Added revenue / year = recovered admissions x \$1,928 x 1.5 x 12. Billing-capture uplift is excluded.

- Home care referral fees = declines x share placed in home care x fee x 12 (private pay only).
- Defaults: 200 referrals/month, 35% acceptance, 45% eligible, 15% recovered, 20% placed, \$300 fee; about \$305K/year added revenue. All adjustable.

5. Synthetic data: how it was created and sourced

All patient-level data was written by hand for this demo (drafted with Claude and reviewed by the team). Nothing was copied from real records. Every facility, payer (other than Original Medicare), clinician, agency and PACE program name is fictional and labeled (*synthetic*) in the app. The goal was a small set of referrals where each one exercises a different branch of the system, with realistic clinical language and deliberately planted gaps.

5.1 Design principles

- **One scenario per branch:** accept, fixable, and three different decline reasons (staffing, clinical, insurance).
- **Planted billing gaps:** discharge summaries document comorbidities that the short nurse summary omits, mirroring the real problem that coders see only the nurse's summary.
- **Realistic structure:** discharge summaries follow common sections (principal diagnosis, hospital course, past history, functional status, disposition, home health orders, face-to-face encounter, follow-up).
- **Homebound and eligibility language** is written the way Medicare documentation typically phrases it (e.g. 'leaving home requires a considerable and taxing effort').
- **Consistent time frame:** all discharges on 26 Sep 2026, so the 48-hour start-of-care deadline is meaningful against the roster's open slots.
- **Real geography, fake people:** San Francisco Bay Area zip codes with approximate real centroid coordinates, so distances are plausible.

5.2 The five referrals

ID	Patient (synthetic)	Scenario	Planted features	Expected outcome
HH-1001	Eleanor Park, 78F, 94110, Original Medicare	Hip fracture after fall, s/p ORIF (hospital)	Nurse summary omits type 2 diabetes with neuropathy on insulin, CKD 3b, recurrent depression; HTN + CKD coding convention	ACCEPT; RN, PT and OT matched; 5 missed diagnoses
HH-1002	Walter Brooks, 82M, 94601, Medicare Advantage (synthetic)	HFpEF exacerbation (hospital)	Prior auth required; face-to-face missing; nurse summary omits COPD, AFib on apixaban, stage 2 sacral pressure injury, obesity	FIXABLE; face-to-face and prior auth drafts; 6-7 missed diagnoses
HH-1003	Rosa Delgado, 86F, 95020 Gilroy, Original Medicare + Medi-Cal	Stage 3 heel ulcer on wound vac, moderate Alzheimer's (SNF)	Only wound-vac-trained RN within radius is first free 73 hours after discharge	DECLINE (staffing); IHSS + MSSP in discharge-planner note; skilled hand-off warning
HH-1004	Harold Jensen, 79M, 94501, Original Medicare; Vietnam veteran; lives alone	Pneumonia, resolved (hospital)	Independent 300 ft, explicitly not homebound; needs help bathing, meals, housekeeping	DECLINE (clinical); VA Aid & Attendance; 2 private-pay agencies
HH-1005	Linda Chen, 62F, 94112, employer PPO (synthetic, out of network); LTC policy	Right MCA stroke, left hemiparesis, dysphagia (hospital)	Payer not contracted, although the agency could staff every discipline	DECLINE (insurance); LTC insurance + PACE; skilled hand-off warning

5.3 Supporting data files

File	Contents	How it was made
nurses.csv	9 clinicians: license, skills, home zip, travel radius, next open slot, visits left	Hand-authored so each referral has a clear staffing result (e.g. a wound-vac RN 9.8 miles away but outside the 48-hour window).
homecare_agencies.csv	4 private-pay agencies: zips served, hourly rate (\$34-40), LTC acceptance, languages	Fictional; rates chosen to be in a plausible Bay Area range for illustration only.
reference.json: payers	Original Medicare, a synthetic Medicare Advantage plan, a synthetic employer PPO	Contract and prior-auth flags set to create the fixable and out-of-network scenarios.
reference.json: zips	10 Bay Area zip codes with approximate centroid lat/lon, county and state	Approximate public zip centroids; used for distance, county and state program lookup.
state_programs.json	Medicaid in-home programs for CA, TX, FL, NY, IL, PA plus a default	Researched summaries of each state's programs (team); simplified eligibility flags, not official determinations.
comorbidity_subgroups.json	12 ICD-10 prefix to PDGM subgroup mappings, 1 interaction pair	Hand-curated subset relevant to the demo referrals; not validated against the current CMS table.
reference.json: PACE areas	Two fictional PACE programs mapped to zip lists	Illustrative; real PACE service areas differ and must be looked up.
referrals.json: manual baseline	Per referral: manual decision, minutes to decision, note	Assumed values (150-240 minutes) representing today's manual process; not measured data.
data/mock/*.json	Cached extraction and billing outputs per referral	Written by hand to match the expected model output; every quote copied verbatim from the synthetic summaries and verified by code.

5.4 Clinical coding notes

ICD-10 codes in the synthetic data and cached outputs follow standard home health conventions (e.g. fracture aftercare with 7th character D, stroke sequela I69.x, diabetes-CKD and hypertension-CKD/heart failure 'with' presumptions). They are illustrative and must be reviewed by a certified coder before any real use.

6. Testing and validation

- **Rules in cached mode:** all five referrals produce the expected decision and reason; every cached quote verifies against its source text.
- **UI:** Streamlit AppTest renders every tab and every referral with no exceptions.
- **Live model (GPT-5.6 Terra on Bedrock):** all five decisions matched the design. Per-referral latency about 8-24 seconds; parallel run of all referrals about 20 seconds (down from about 75 seconds sequentially).
- **Discharge-planner notes:** live notes for all three declines list home care options, the skilled hand-off where needed, and the family's preferred language.
- **Guardrail observed:** in one live run the model paraphrased an evidence quote; quote verification flagged it in the UI.

6.1 Fixes made after live testing

Issue found	Fix
Billing step listed the primary diagnosis as 'missed' although the nurse summary mentioned it by abbreviation (CHF, hip ORIF)	Prompt now excludes conditions already mentioned, even by abbreviation, unless the code should change
Early versions texted families a care plan	Replaced with a note to the hospital discharge planner on the referral file; agencies may not contact declined patients
Overview cited 77% staffing and ~30%/~80% acceptance from a vendor blog	Replaced with WellSky first-party data (~24% from ~46%); 77% removed as a citation error
Overview claimed agencies decline 'often without checking the real roster' (unsourced)	Removed; card now cites the 48-hour rule in 42 CFR 484.55
21% first-responder figure had no source	Replaced with WellSky's 22% (company-reported), flagged for pilot verification
Run-all took about 75 seconds	Parallelized with a 5-thread pool
Dollar signs rendered as LaTeX math in Streamlit captions	Dollar signs escaped in captions and model-generated notes
An unsourced '20-40 page packet' claim on the Overview tab	Removed; replaced with 'long discharge packets'

7. Guardrails, compliance and security

- **No real PHI:** synthetic data only. A production version needs a BAA with AWS, Bedrock data-handling review and HIPAA controls.
- **Clinical safety:** Oikos accepts only when a real clinician slot exists inside the 48-hour start-of-care window; declined skilled patients trigger a hand-off warning.
- **Coding integrity:** suggestions show chart quotes and require clinician confirmation; nothing is auto-coded or auto-billed.
- **Anti-kickback:** no payments tied to hospital referrals; flat software pricing; home care referral fees only for private pay, pending healthcare counsel review. IHSS is consumer-directed, so there is usually no agency to charge.
- **Eligibility language:** 'likely' / 'possible' only; the county, VA or insurer decides.
- **Credentials:** AWS keys are never stored in the repo; locally they live in a temporary session file; on Streamlit Cloud they would go in Secrets. The public link runs cached mode by default so no one can spend AWS credits.

8. Risks and open questions

Risk / question	Why it matters	Next step
CarePort automation and terms of service	No API; the agent depends on agency credentials and could be blocked	Review terms; explore official partner or EHR routes
22% first-responder statistic	Company-reported by WellSky with no published methodology	Measure placement rate vs response time with a pilot agency
Acceptance data from a single network	WellSky/CarePort data may not represent all agencies	Cross-check with claims-based studies (e.g. referral fulfillment rates)
State program rules	Six states summarized; rules change and differ by county	Source from official state Medicaid pages and maintain them
PDGM estimate is a subset	12 of CMS's subgroups and 1 interaction pair; weights change yearly	Load the full annual CMS subgroup table
Real roster and scheduling data	Matching is only as good as the schedule data	Integrate with agency EMRs (e.g. WellSky, HCHB)

9. Roadmap

Phase	Scope
Now (hackathon)	Triage + billing capture + home care qualification on synthetic data; public demo link
Next	Pilot with 1-3 California agencies; real portal ingestion (CarePort, fax OCR, Epic); EMR roster sync; full CMS PDGM table
Later	SNF/LTC referral sourcing; discharge-planner tools; Medicare Advantage / ACO contracts; more states; standalone coding platform

Appendix: running the demo

```
cd "09 Hackathons/aws/oikos"
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
streamlit run app.py
```

```
# live model: export AWS workshop credentials in the same terminal first
export AWS_REGION=us-east-1 AWS_ACCESS_KEY_ID=... AWS_SECRET_ACCESS_KEY=... AWS_SESSION_TOKEN=...
```

Sources: MedPAC home health update (Dec 2025); WellSky 2023 Evolution of Care Report via Home Health Care News (Jul 2023); Home Health Care News on WellSky referral data (Dec 2025); WellSky press release on referral intake (Mar 2026); 42 CFR 484.55; CMS CY2026 Home Health PPS final rule (CMS-1828-F); Hospice News and STAT on the UnitedHealth-Amedisys transaction. Program rules are general summaries and must be verified with official sources.